

SQL Functions

- SQL Functions are built-in operations that perform calculations or transformations on data.
- They help in summarizing data, formatting results, handling NULL values, and extracting useful information.

Types of SQL Functions

SQL Functions are mainly divided into:

- Single Row (Scalar) Functions
- Aggregate (Group) Functions

1. Single Row (Scalar) Functions

These functions work on each row individually and return one result per row.

A) String/Character Functions

- **UPPER()**: Converts text to uppercase

Example: `SELECT UPPER('hello');` → HELLO

- **LOWER()**: Converts text to lowercase

Example: `SELECT LOWER('HELLO');` → hello

- **LENGTH()**: Returns length of a string

Example: `SELECT LENGTH('Database');` → 8

- **CONCAT():** Joins two or more strings

Example: **SELECT CONCAT('SQL','Functions'); → SQLFunctions**

- **SUBSTRING():** Extracts part of a string

Example: **SELECT SUBSTRING('Computer',1,4); → Comp**

B)Numeric Functions

ABS(): Returns absolute value

Example: **SELECT ABS(-20); → 20**

ROUND(): Rounds off a number

Example: **SELECT ROUND(12.678,2); → 12.68**

MOD(): Returns remainder

Example: **SELECT MOD(10,3); → 1**

C) Date & Time Functions

CURRENT_DATE: Returns today's date

NOW(): Returns current date and time

YEAR(): Extracts year from a date

Example: **SELECT YEAR('2026-02-08'); → 2026**

D) Conversion Functions

CAST(): Converts one datatype to another

Example: **SELECT CAST('100' AS INT); → 100**

E) NULL Handling Functions

IFNULL(): Replaces NULL with a given value

Example: **SELECT IFNULL(NULL,0); → 0**

F) FORMAT() Function (New)

FORMAT(): Formats a value or date in a readable form.

Example: **SELECT FORMAT(12345.678,2); → 12,345.68**

2 Aggregate (Group) Functions

Aggregate functions work on multiple rows and return a single result.

COUNT(): Counts number of rows

Example: **SELECT COUNT(*) FROM Student;**

SUM(): Returns total sum

Example: **SELECT SUM(Salary) FROM Employee;**

AVG(): Returns average value

Example: **SELECT AVG(Marks) FROM Student;**

MAX(): Returns maximum value

Example: **SELECT MAX(Salary) FROM Employee;**

MIN(): Returns minimum value

Example: **SELECT MIN(Marks) FROM Employee;**

Example Table: Employee

Dept	Salary	Employee
IT	50000	A
IT	60000	B
HR	40000	C

Query Example with GROUP BY:

```
SELECT Dept, AVG(Salary)
FROM Employee
GROUP BY Dept;
```

Output:

Dept	AVG(Salary)
IT	55000
HR	40000

Where SQL Functions Can Be Used

SQL Functions can be used in:

- SELECT clause
- WHERE clause
- GROUP BY clause
- ORDER BY clause

Feature	Scalar Function	Aggregate Function
Works on	Each row	Multiple rows
Output	One result per row	One result per group
Examples	UPPER(), ROUND()	SUM(), AVG()